

What's Behind the Rural Gap in Electric Vehicle Adoption? Investigating Needs, Barriers, Beliefs, and Values in Vermont and Maine



University of Vermont
Transportation Research Center

Nazmus Sakib (nsakib@uvm.edu), Dana Rowangould (drowango@uvm.edu), Elizabeth Doran (elizabeth.doran@uvm.edu)

Motivation

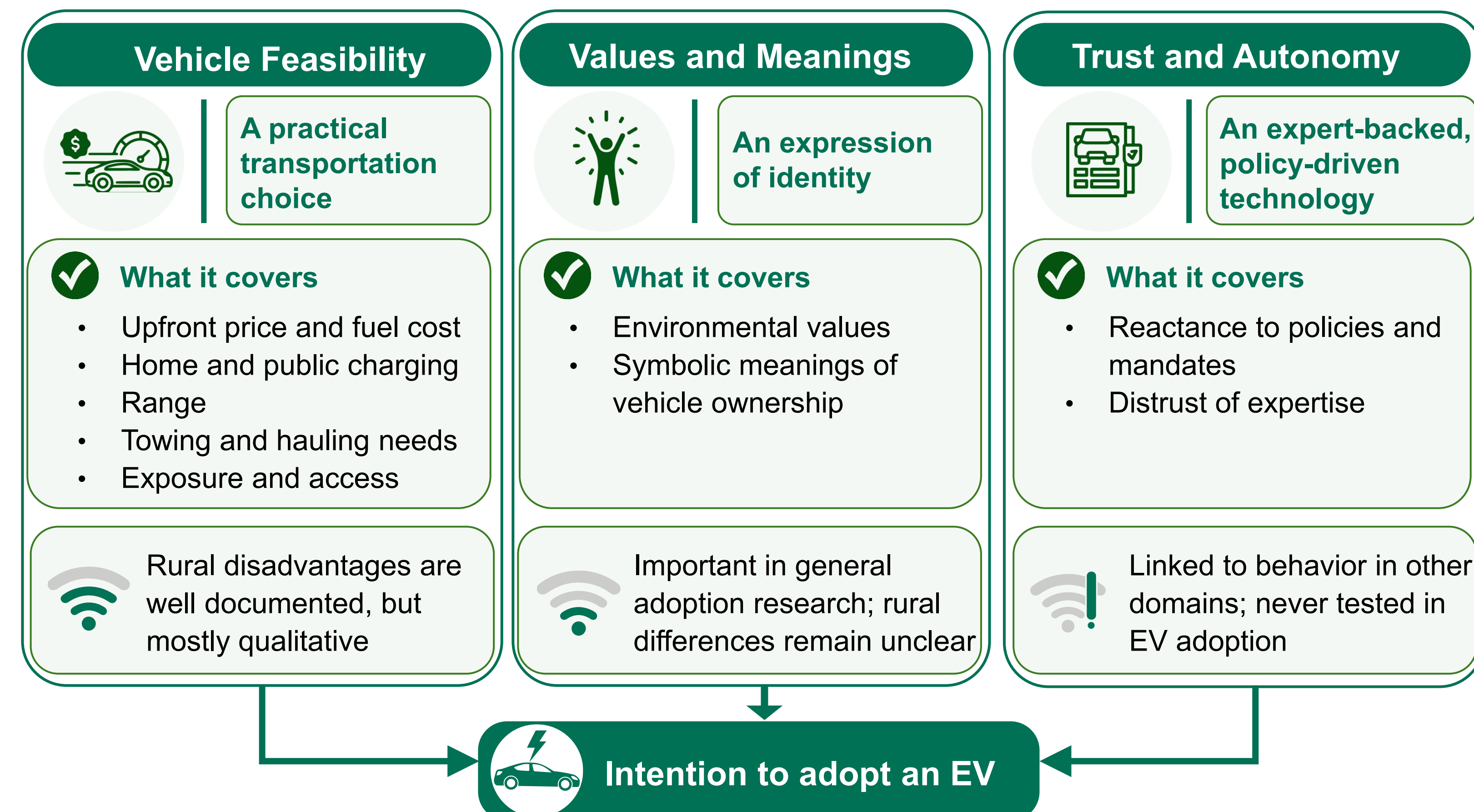
40%

lower
rural EV
adoption

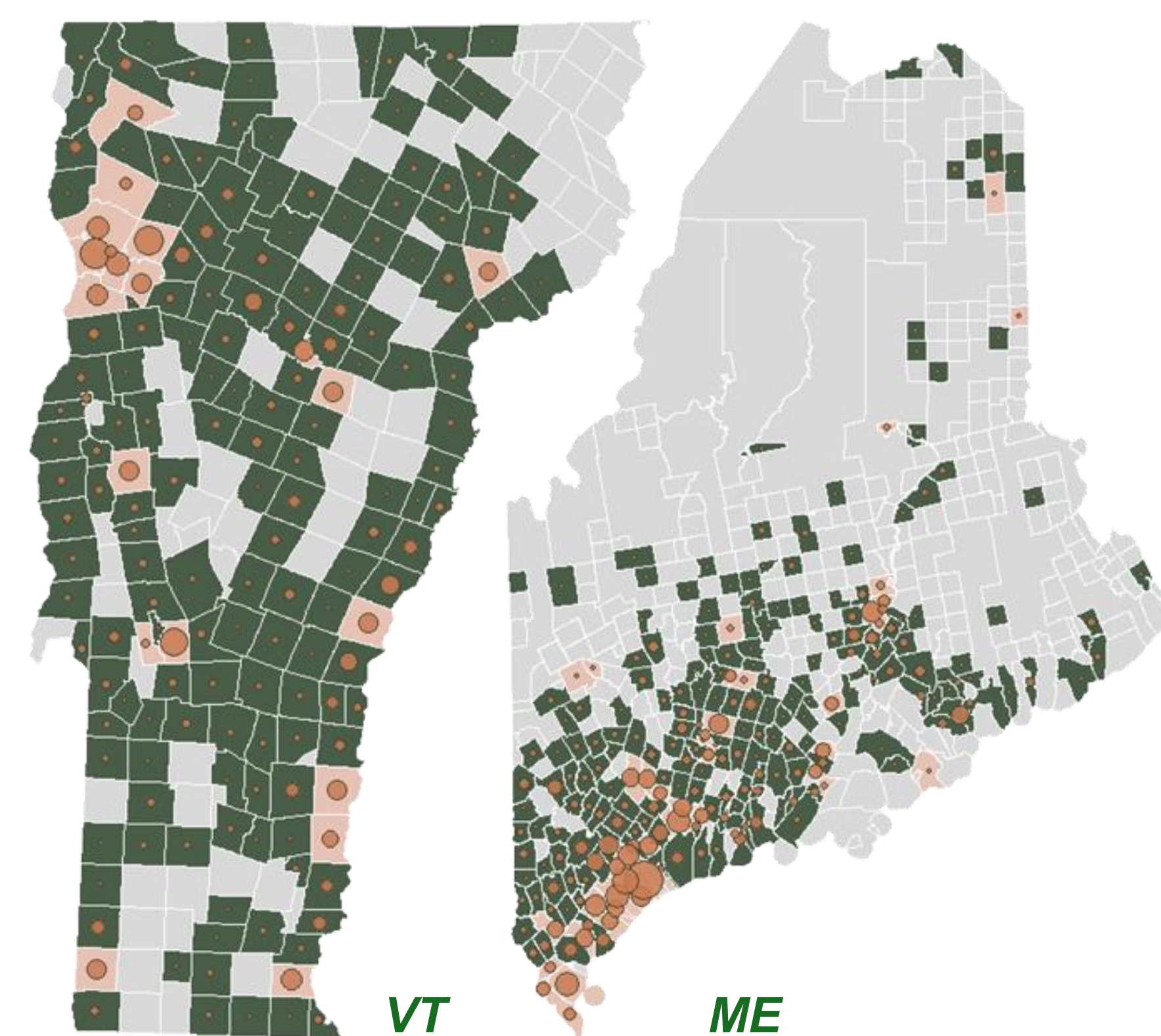
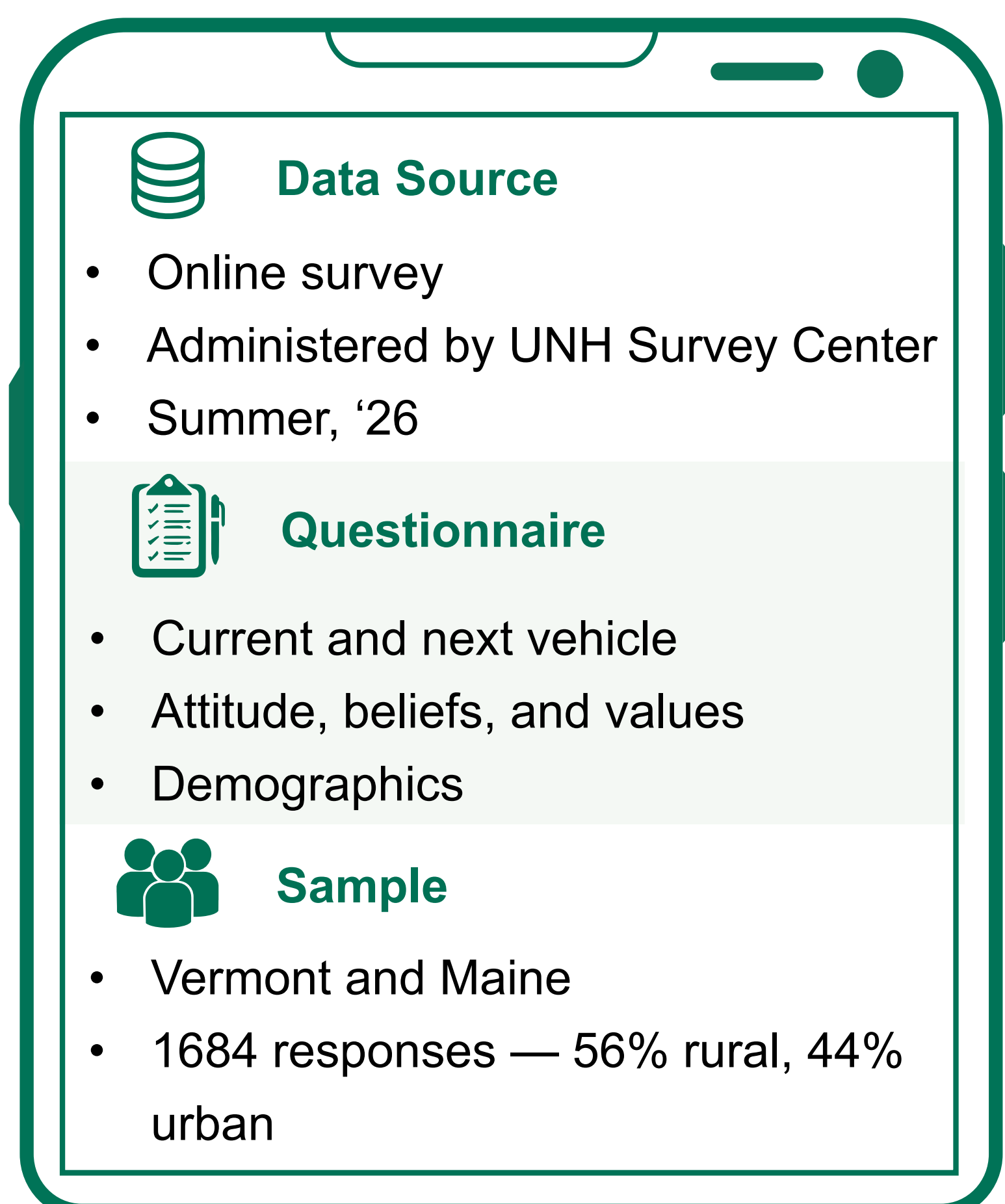
- Electric vehicles (EVs) are a central lever for state climate goals.
- Adoption is markedly lower in rural areas than in urban ones.
- The gap has persisted even as EVs have become more common overall.

Research Gap and Framework

Three accounts could explain the rural adoption gap. They differ in how much we know.



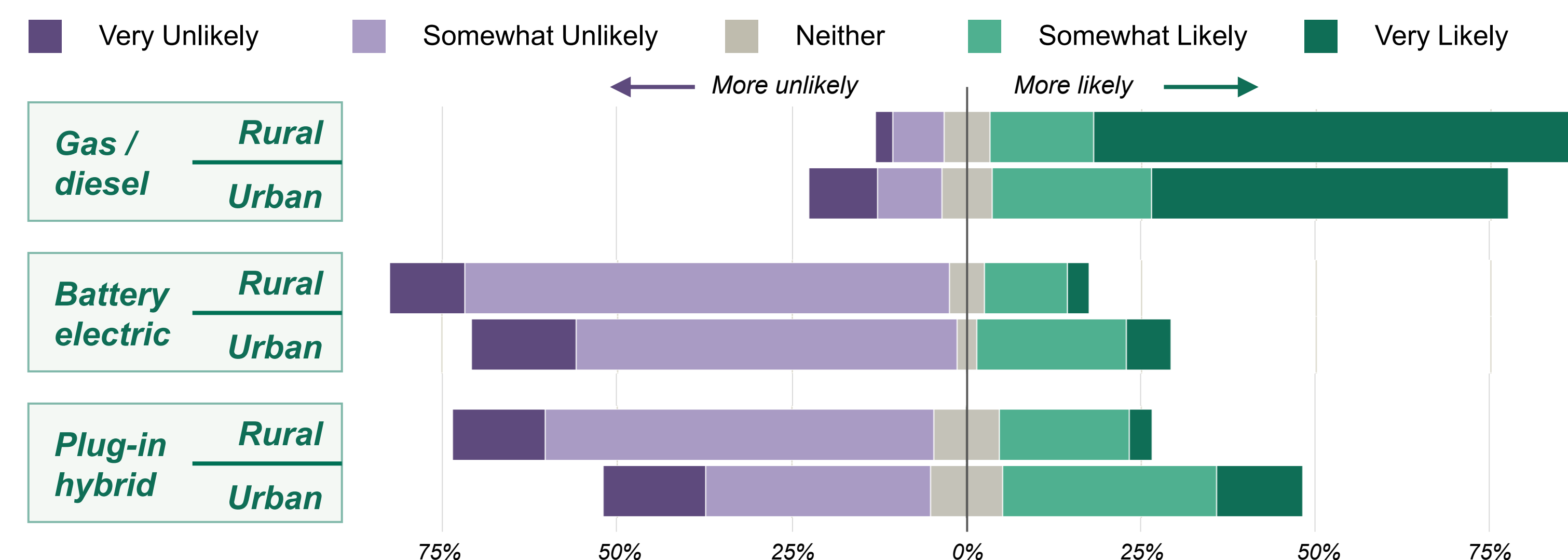
Study and Sample



■ Rural ■ Urban ● Proportion of respondents

Preliminary Results

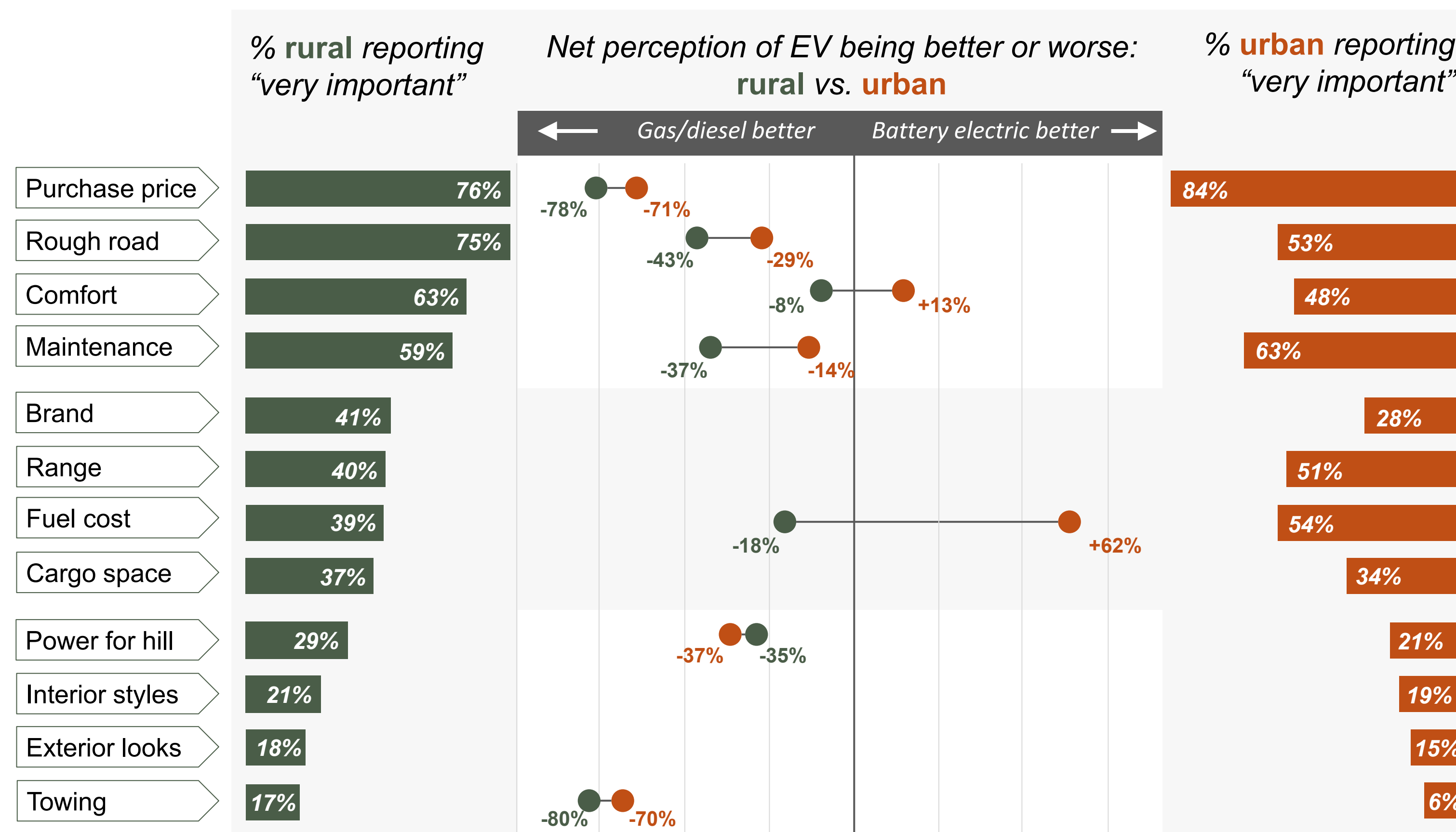
How likely is your next vehicle to be...



Takeaway

- Rural respondents are more likely to reject a battery-electric or plug-in hybrid next; urban respondents are more open to the plug-in hybrid.

How important is each feature in your next vehicle purchase, and how does a battery electric vehicle compare to gas?



Takeaway

- Rural respondents rate battery-electrics worse than gas on nearly every attribute — power for hills being the only exception.
- Fuel savings are EVs' strongest case, yet rural drivers still favor gas — the widest rural–urban gap here.

Contributions

Theoretical contributions

- When complete, it will be the first quantitative US test of the rural EV gap — past work is thin and qualitative.
- Will be the first to test trust and autonomy as an EV barrier.

Practical contributions

- Will be able to show if the gap is a cost, identity, or trust problem — each needs a different fix.
- Will be able to flag where mandates may backfire on distrustful drivers.

Next Steps

- Move from early findings to a full rural–urban model of what drives the gap
- Full results expected in November '26

Acknowledgements

The author thanks Dr. Gregory Rowangould (UVM, PI), Dr. Jonathan Rubin (UMaine, co-PI), and Dr. Trisha Shrum (thesis committee chair) for their guidance, and the UNH Survey Center for survey administration. This material is based on work supported by the National Science Foundation.